Floating-Point numbers

$$x \in \mathbb{F} \subset \mathbb{R}$$

$$x = (-1)^s \cdot \underbrace{(0.a_1 a_2 a_3 \cdots a_t)}_{\text{mantissa}} \cdot \beta^e$$

digits

$$\mathbb{F} = \mathbb{F}(\beta, t, L, U)$$

basis

number of significant digits

IEEE-754 64 bit (double precision)

$\beta = 2$
$t = 52$
$L = -1021$
$U = +1024$

<u>Example</u> $\quad x \in \mathbb{R} \qquad fl(x) \in \mathbb{F}(\beta, t, L, U)$

$$\beta = 10, \; t = 4, \; L = -5, \; U = +5$$

$$x = 131.76421737$$

$fl(x) = 0.1317$ (truncation) or 0.1318 (rounding)

Absolute error : $|x - fl(x)|$

Relative error : $\dfrac{|x - fl(x)|}{|x|}$ $\quad |x| \neq 0$

Uniformly bounded relative error

$$\frac{|x - fl(x)|}{|x|} \leq \frac{1}{2} \varepsilon_M \qquad x_{min} \leq x \leq x_{max}$$

$$x_{min} \simeq 2.22 \times \underline{\underline{10^{-308}}} \qquad x_{MAX} \simeq 1.79 \times \underline{\underline{10^{308}}}$$

$$\varepsilon_M = \min_{x \in \mathbb{F}} \{ x \mid 1 + x > 1 ,\ x > 0 \}$$

$$x = (-1)^s (0{,}a_1 a_2 a_3 a_4 \cdots a_t) \cdot \beta^e$$

$$a_1 \neq 0$$

$$1 = (-1)^0 (0{,}10000 \cdots 0) \cdot \beta^1$$

$$1 + \varepsilon_M = (-1)^0 (0{,}100 \cdots 1) \cdot \beta^1$$

$$\varepsilon_M = (1 + \varepsilon_M) - 1 = (-1)^0 (0{,}000 \cdots 1) \cdot \beta^1$$

$$\underline{\varepsilon_M} = \beta^{-t} \cdot \beta^{1} = \beta^{1-t} \simeq 2^{-52}$$

in IEE754

because I can store 53 digits with 52 bits given the normalization condition

Example

$a = 1.0 \times 10^{+308}$ $b = 1.110 \times 10^{+308}$ $c = -1.001 \times 10^{+308}$

$$a + (b + c) = (a + b) + c$$

$$a + b > x_{MAX} \xRightarrow{\text{overflow}} \text{Inf}$$

$$\text{Inf} + c \Rightarrow \text{Inf}$$

$$a + (b + c) = 1.099 \cdot 10^{+308} < x_{MAX}$$

Loss of significant digits

(Numerical cancellation)

$x \neq 0$ $$y = \frac{((1 + x) - 1)}{x} = 1$$

$$\hat{y} = \frac{((1 + fl(x)) - 1)}{f(x)} = 0 \qquad x > x_{MIN}$$

if $|x| < \varepsilon_M$

$$fl(x) < \varepsilon_M$$

$$\tilde{y} = \frac{((1-1)+fl(x))}{f(x)} = 1$$

Summary :

- Floating point numbers are **NOT** real numbers

- Arithmetics with F.P. has slightly different rules

- Approximating real numbers with F.P. causes a relative error which is uniformly bounded by ε_M

- Numerical algorithms may amplify errors on input data

- Rule of thumbs : NEVER check for equality between F.P. numbers in your algorithm, i.e.

$$y == x \quad \text{BAD}$$

$$|y - x| \leq \tau \quad \text{GOOD}$$

Solving nonlinear equations

→ QSG : 2.2 – 2.3 – 2.6 3rd ed.

→ QSS : 6.2 – 6.3 – 6.5 3rd ed.

- Find zeros/roots of a smooth function

$$f : \mathbb{R} \supset [a, b] \longrightarrow \mathbb{R}$$

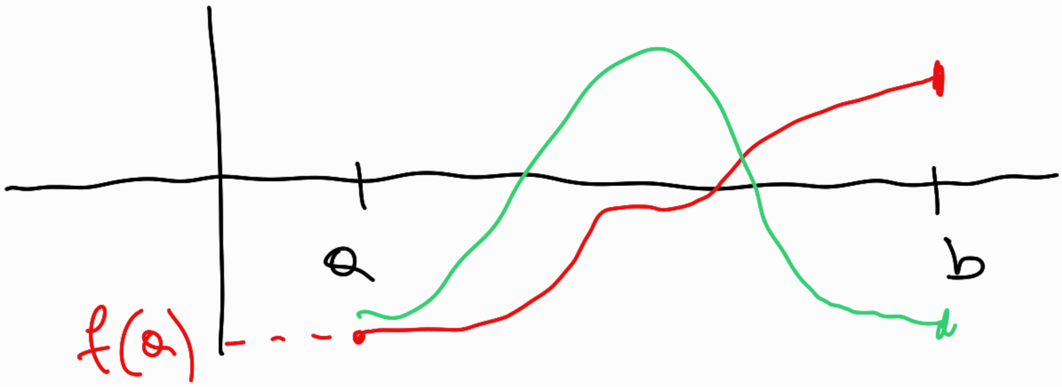
$$f \in C^0([a, b])$$

Find $\xi \in [a, b]$ s.t. $f(\xi) = 0$

- Existence is guaranteed by the Bolzano theorem

$$\left.\begin{array}{l} f \in C^0([a, b]) \\ f(a) \cdot f(b) < 0 \end{array}\right\} \Rightarrow \exists\, \xi \in [a, b] \text{ s.t. } f(\xi) = 0$$

Sufficient condition



Bisection algorithm

$a_0 = a \qquad b_0 = b \qquad x_0 = \dfrac{a_0 + b_0}{2}$

Initialization $\quad \xi \simeq x_0 \qquad \varepsilon_0 = |\xi - x_0| = \dfrac{L}{2}$

$L = \dfrac{b_0 - a_0}{2}$

Iteration $\quad$ for $i = 1, 2, 3, \dots$

- if $|a_{i-1} - b_{i-1}| < 2\varepsilon$ tolerance
- else

  if $f(x_{i-1}) == 0 \Rightarrow \xi = x_{i-1}$

  else if $f(x_{i-1}) \cdot \underline{f(a_{i-1})} < 0$

  $a_i = a_{i-1}$

  $b_i = x_{i-1}$

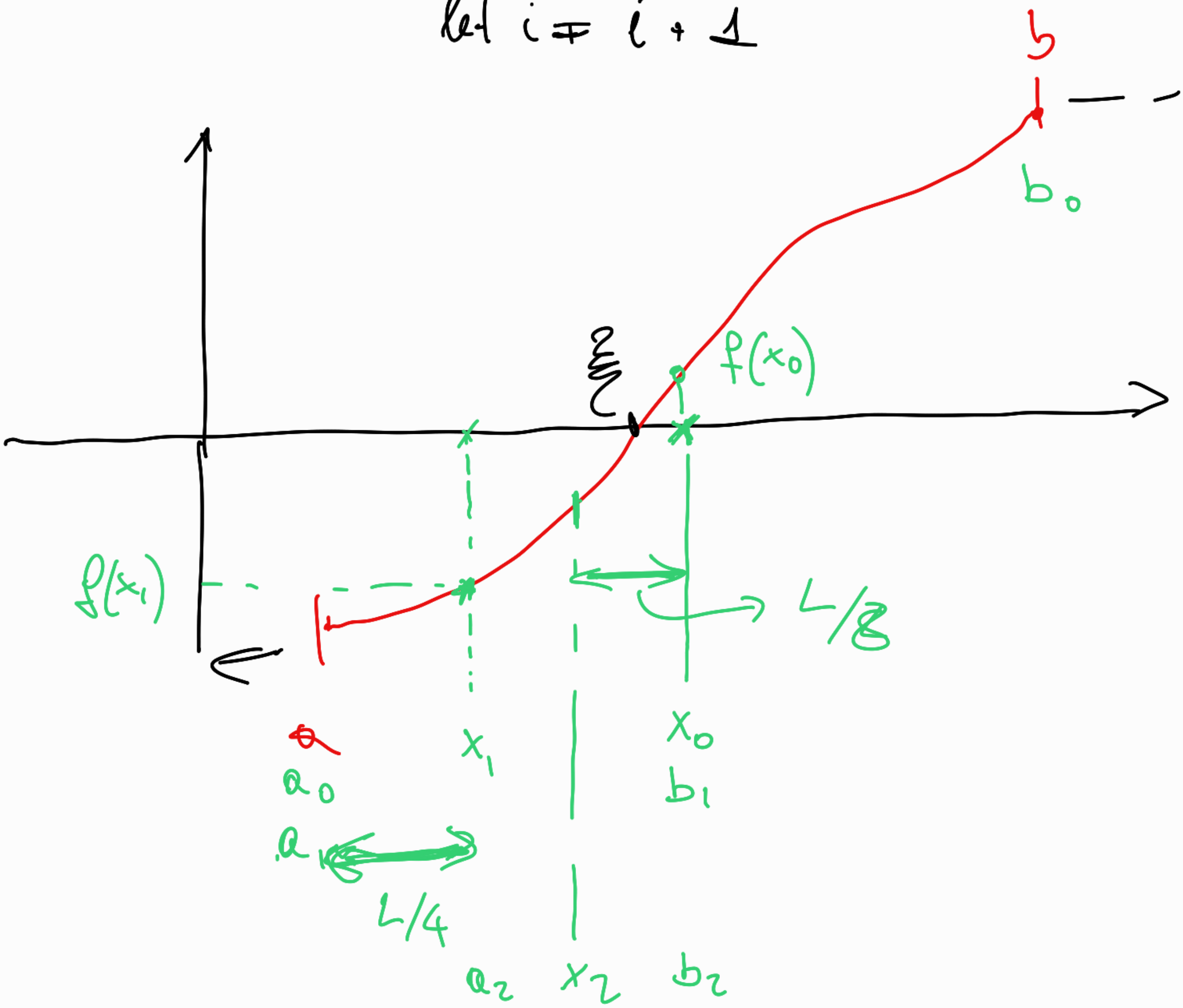
  $x_i = \dfrac{a_i + b_i}{2}$

else if $f(x_{i-1}) \cdot f(b_{i-1}) < 0$

$$a_i = x_{i-1}$$
$$b_i = b_{i-1}$$

$$x_i = \frac{a_i + b_i}{2}$$

let $i = i + 1$



After i steps of the iteration

$$a_i < \xi < b_i \qquad x_i = \frac{a_i + b_i}{2}$$

$$\varepsilon_i = |\xi - x_i| \leq \frac{b_i - a_i}{2}$$

$$L = b - a \qquad L_i = b_i - a_i$$

$$L_i = 2^{-i} L \qquad \varepsilon_i \leq \frac{L_i}{2}$$

In order to have We can require

$$\varepsilon_N \leq \tau \quad \Leftarrow \quad \varepsilon_N \leq \frac{1}{2} L_N \leq \tau$$

$$2^{-N-1} \leq \frac{\tau}{L} \quad \Leftarrow \quad N + 1 \geq \log_2 \frac{L}{\tau}$$

If $N \geq \log_2 \frac{L}{\tau} - 1$ then $\varepsilon_N \leq \tau$

<u>a-priori estimate</u>